When Do Geometric Markers Predict Generalization? A Controlled Analysis of Task-Relevance and Shift Type

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Paper under double-blind review

Abstract

Neural representation geometry has recently been proposed as a cheap, training-free proxy for predicting a model's generalization performance. Recent work (Chou et al., 2026) shows that geometry-derived markers such as the effective dimension (D_{eff}) correlate strongly with out-of-distribution (OOD) accuracy and outperform in-distribution (ID) test accuracy as a model-selection criterion. This paper audits that *comparison* rather than competing with it. We argue that such comparisons require three controls that are rarely applied: (i) they must be made within a subset of models with near-identical ID accuracy, since ID accuracy is the very quantity the markers are meant to supplement; (ii) architecture main effects must be removed, since architecture families differ systematically in both ID accuracy and representational geometry; and (iii) marker strength must be reported with rank-based statistics alongside Pearson correlations, since a strong Pearson coefficient alone can be driven by a few extreme points. We formalize these as a four-condition protocol — ID matching, group residualization, dual correlation statistics, and cross-group sign consistency — and report what it reveals when applied to the per-model measurements released with the original study (5 architectures, 224 models: each architecture trained on its own 4 learning rates × 4 weight decays × 2–3 seeds, with the union of hyperparameter values spanning 13 learning rates and 9 weight decays), to a second release of that study (80 models, 6 corruption scenarios), and to 80 models we trained ourselves across three architectures, of which 79 are healthy after excluding one diverged run. Three findings emerge. (i) **The task-relevant markers survive all four controls.** D_{eff} retains a strong, rank-consistent, sign-consistent association with OOD accuracy under the ID-matched, architecture-adjusted protocol ($r = +0.93$, $\rho = +0.85$, $p < 10^{-4}$, positive in all five architectures), with 100% bootstrap robustness. This is not a Simpson's paradox. (ii) **The accuracy baseline is not a stable reference point.** At fixed ID accuracy with architecture effects removed, the sign of the ID-accuracy association is not consistent across populations: on the two releases of the original study it is positive on one ($+0.39$, 95% CI $[+0.22, +0.62]$) and negative on the other (-0.63, 95% CI $[-0.83, -0.42]$), and on our own 48 models it is negative in aggregate but *flips sign within training families*, failing the protocol's own sign-consistency condition. An accuracy baseline whose sign depends on the population cannot anchor a "markers beat accuracy" claim in either direction. We localize the cause: on the 224-model release the positive sign is mediated by the learning-rate axis — both accuracies track architecture-adjusted learning rate in the same direction, and blocking that pathway collapses the association from $+0.39$ to $+0.04$ — whereas the 80-model release's negative sign survives blocking ($-0.63 \to -0.67$). The two releases therefore differ in confounding structure, not merely in correlation. (iii) **The markers' advantage is conditional on the shift type.** Under photometric and structural corruptions (fog, brightness, contrast), ID accuracy is restored as the strongest predictor ($\rho_{\text{resid}} = +0.82$ to $+0.97$, ranked 1st of 12 markers); under noise and compression corruptions it weakens ($\rho_{\text{resid}} = -0.11$ to $+0.19$, ranked 2nd–7th), and no single candidate marker dominates it there. A single aggregate statement about corruption robustness aver-

Use of LLMs: Large language models were used as writing and coding assistants in preparing this manuscript; all scientific content, analyses, and reported numbers were determined and verified by the authors. A detailed statement appears in the Use of AI tools section.

ages over two opposite regimes. A third, independent observation explains why the baseline behaves this way. On our 48 self-trained models, *task-agnostic* geometric markers and ID accuracy are **nearly uncorrelated with each other** (-0.17 to -0.04) while the markers correlate at $+0.90$ to $+0.95$ with OOD accuracy and ID accuracy does not (-0.05). The two candidate signals measure different things — how well the network fits the training distribution versus how its learned representation is organised — and on class-level shifts only the second is informative. We conclude that the predictive value of representational-geometry markers is conditional on both the task-relevance of the marker and the type of distribution shift, and that claims that geometric markers "replace" ID accuracy must state these qualifications explicitly. We release the control protocol as a reusable diagnostic and validate it on independently trained model families across two class-level shifts, where the baseline's sign instability replicates.

1 Introduction

Predicting how a trained network will behave beyond its training distribution is a long-standing problem with growing practical stakes. Test accuracy on held-out in-distribution (ID) data is the default summary of model quality, but it is exactly the quantity that fails to anticipate out-of-distribution (OOD) behavior. A promising alternative has emerged from the study of representational geometry: rather than evaluating a model on data, one measures geometric properties of its internal representations — class manifolds, their dimensionality, radius, and alignment — and asks whether these quantities track (or even forecast) downstream performance.

Two families of geometric markers have been explored. The first uses task-agnostic statistics of the feature distribution: participation ratio, mean pairwise distances and angles, covariance sparsity. The second uses task-relevant quantities derived from manifold capacity theory: the GLUE framework (Chou et al., 2025) defines, for each class manifold, an effective dimension D_{eff}, effective radius R_{eff}, and effective utility Ψ_{eff}, computed from anchor points of a quadratic program that mirrors linear separability. Recent work by Chou et al. (2026) — hereafter "the original study" — reports that these task-relevant markers, measured on ID data, forecast OOD generalization failures, and do so more reliably than ID test accuracy itself.

The comparison at the heart of that claim — geometric markers versus ID accuracy — is the subject of this paper. We show that as stated, the comparison is not yet well defined, and that once it is made well defined, both parts of the claim change: the markers' advantage grows, but so does the set of conditions under which it fails.

Three unhandled confounds. Comparisons of a proposed proxy against an accuracy baseline are typically made on a pooled correlation over a heterogeneous model population: five architectures, each trained under multiple hyperparameter settings. This design leaves three confounds untreated. First, ID accuracy is not controlled: if two architecture families differ systematically in both ID accuracy and OOD transfer, the pooled correlation between ID accuracy and OOD performance is dominated by between-architecture differences, not by the within-architecture relationship that model selection actually relies on. Second, no per-group sign consistency is checked: a pooled coefficient can be positive while every within-architecture coefficient is negative — the textbook setting of Simpson's paradox. Third, shift type is not treated as a variable: "OOD" spans class-level shifts (new object categories) and corruption shifts (photometric and structural degradations), and there is no a priori reason the same marker should dominate under both.

What we find. We re-analyze the per-model measurements released with the original study — 224 models across five architectures, each architecture trained on its own 4×4 learning-rate/weight-decay grid with 2–3 seeds (the union of hyperparameter values spans 13 learning rates and 9 weight decays; this is a union of per-architecture grids, not a Cartesian product), including the study's own released values of D_{eff}, R_{eff}, and Ψ_{eff} — under a protocol that (a) restricts attention to models with near-identical ID accuracy (± 1 percentage point), (b) residualizes out architecture main effects, (c) reports Pearson and Spearman coefficients jointly and requires them to agree, and (d) requires the sign of each association to be consistent across architectures. We then apply the identical protocol to the study's second release (80 models, 6 corruption scenarios) and

to 80 models we trained ourselves across three architectures (79 healthy) and two class-level shifts. Three findings emerge.

1. *The geometric markers survive the controls.* Within the ID-matched, architecture-adjusted subset ($n = 106$), D_{eff} remains strongly associated with OOD accuracy ($r = +0.93$, $\rho = +0.85$, both $p < 10^{-4}$), with consistent sign in all five architectures and 100% bootstrap robustness. The effect is real, not an aggregation artifact.

2. *The baseline is worse than weak — it is not a stable reference point.* On the 224-model release, ID accuracy has essentially no pooled association with OOD accuracy ($r = -0.05$); after architecture adjustment its Pearson coefficient is $+0.39$. On the study's second release (80 models, 6 corruption scenarios), the same protocol yields $r = -0.63$. Stratified bootstrap over the full pipeline gives 95% confidence intervals $[+0.22, +0.62]$ and $[-0.83, -0.42]$ on the two datasets respectively: both exclude zero, in opposite directions. On our own 48 models the aggregate is negative (-0.44) but the sign *flips within training families*, so the baseline fails the protocol's own sign-consistency condition there. More fundamentally, on those same 48 models the baseline is *nearly uncorrelated with the geometric markers* (-0.05 with OOD accuracy, -0.17 with marker values), so the two candidate signals are not competing measurements of a common quantity — they are largely independent, and only one of them tracks class-level transfer. An ID-accuracy baseline whose sign depends on the population cannot anchor a "markers beat the baseline" claim in either direction.

3. *The markers' advantage is shift-type dependent.* Under photometric and structural corruptions (fog, brightness, contrast), ID accuracy is again the strongest predictor of corruption accuracy (ρ_{resid} up to $+0.97$, 1st of 12); under noise and compression corruptions it weakens (ρ_{resid} between -0.11 and $+0.19$, ranked 2nd–7th), and geometry-derived markers take over. The original study reports the corruption case as a single aggregate; splitting it reveals that both halves of the aggregate statement hold for disjoint reasons. We report this decomposition under both an uncontrolled and a fully controlled instrument, and the split survives either way.

Independent validation. Because all of the above analyzes the original authors' own released measurements, we additionally train 48 ResNet-18 models under a 4×4 learning-rate/weight-decay grid with 3 seeds, plus 16 models each of ShuffleNetV2-x1.0 (Ma et al., 2018) and RegNetY-400MF (Radosavovic et al., 2020), and run the identical protocol with publicly computable markers on two independent class-level shifts: CIFAR-100 and Tiny-ImageNet (Le & Yang, 2015) linear probes. The baseline's negative aggregate sign replicates on both shifts (residualized $r = -0.44$ and -0.34), but a per-family sign check reveals that it is *not* uniformly negative — it flips to $+0.69$ in the $lr = 1.0$ family, which is a low-transfer training regime (OOD 7.1–14.7% against 13.7–32.4% for the other three, at comparable ID accuracy). Restricting to the three high-transfer families leaves the baseline negative in all three ($-0.97/-0.60/-0.33$) and strengthens the aggregate to -0.66. An independently implemented version of the GLUE dimension — usable for relative comparisons only — agrees with the released implementation across datasets ($r = +0.91$ vs. $+0.87$), which we report as a cross-implementation robustness check.

Figure 6 in Appendix A is a conceptual schematic of the two-dimensional structure this paper argues for (task-relevance of the marker $\times$ type of the shift); it is a roadmap rather than evidence, and all empirical support is given in §4 onward.

Contributions. This paper is an audit of comparison methodology, not a competing claim that geometric markers predict generalization better than the original study concluded. Our subject is the *comparison* — whether a cheap proxy can be said to beat an accuracy baseline — and our contributions are:

1. A four-condition control protocol (ID-matching, confounder residualization, dual correlation statistics, cross-group sign consistency) for evaluating any proposed "cheap proxy" against an accuracy baseline, with evidence that on each of our three data sources every condition is load-bearing.

2. A controlled re-analysis on the original study's 224-model release showing the geometric markers' effect is not a Simpson's paradox — and a cross-group sign check that the protocol's own machinery can be turned on the *baseline*.

3. The observation that the ID-accuracy baseline has a population-dependent sign — with non-overlapping bootstrap CIs on the original study's two releases, and a within-family sign flip on our own models — which undermines pooled "markers vs. accuracy" comparisons in both directions.

4. A shift-type decomposition of the corruption result, reported under both an uncontrolled and a fully controlled instrument, showing when ID accuracy dominates and when it collapses.

5. Evidence that geometric markers and ID accuracy carry near-orthogonal information on independently trained models (§4.8.4): markers correlate -0.17 to -0.04 with accuracy but $+0.90$ to $+0.95$ with OOD transfer, and the association survives conditioning on accuracy ($+0.69$ to $+0.86$).

6. Independent replication on 80 self-trained models across three architectures and two class-level shifts, plus a cross-implementation check on the GLUE dimension.

2 Related Work

Representational geometry and generalization. Geometric summaries of representations have long been proposed as signals of learning quality. Effective-rank and related rank statistics have been studied as generalization indicators (Jiang et al., 2020); compression-oriented accounts link generalization to the parsimony of internal representations (Sorscher et al., 2022); double-descent phenomena complicate any monotone story about capacity and generalization (Nakkiran et al., 2020). Our contribution to this line is not a new statistic but a protocol: we show that several published-looking comparisons in this space are pooled across heterogeneous model families in ways that can invert the sign of the baseline.

Manifold capacity and GLUE. Manifold capacity theory (Chung et al., 2018) connects the linear separability of object manifolds to effective geometric quantities — radius and dimension — defined via anchor points of a constrained quadratic program. The GLUE framework (Chou et al., 2025) develops this into a practical analysis toolkit and defines the effective utility Ψ_{eff}. The original study (Chou et al., 2026) applies these markers to OOD prognosis and model selection. We take the released measurements of that study as our primary data; our re-analysis is therefore an audit of the comparison methodology, not of the computations.

OOD robustness benchmarks. Corruption robustness benchmarks (Hendrycks & Dietterich, 2019) distinguish corruption types that our analysis shows are not interchangeable as shift types: photometric and structural corruptions behave differently from noise and compression corruptions with respect to which predictor dominates. Class-level transfer evaluations via linear probes follow standard practice; we adopt the original study's protocol (frozen features, new head trained on the shift dataset) for comparability.

Statistical methodology. Our protocol instantiates standard cautions from observational statistics: Simpson's paradox and aggregation bias, confounder adjustment by residualization (Pearl, 2009), rank statistics as robustness checks against leverage points, and stratified bootstrap for pipeline-level uncertainty (Efron & Tibshirani, 1993). The methodological point we add is that for "proxy beats baseline" claims in model evaluation, all four cautions bind simultaneously, and we exhibit a concrete case where omitting any one of them yields a qualitatively wrong conclusion.

Gap. No prior work, to our knowledge, has evaluated geometry-based generalization markers under an ID-matched, architecture-adjusted protocol; nor has the sign stability of the ID-accuracy baseline itself been examined across releases of the same benchmark family.

3 Method

3.1 Preliminaries: markers and notation

For a model f with penultimate features $\phi(x) \in \mathbb{R}^d$, let $\{X_\mu\}_{\mu=1}^P$ denote class-conditional feature sets on ID evaluation data (M samples per class). Task-agnostic markers summarize the pooled or per-class feature distribution: participation ratio $\text{PR} = (\sum_i \lambda_i)^2 / \sum_i \lambda_i^2$ over the eigenvalues of the feature covariance;

numerical rank; NC1 (within-class to between-class covariance ratio); mean pairwise distance, angle, and covariance magnitude; sparsity.

Task-relevant markers come from the GLUE framework: for each direction t sampled from $\mathbb{R}^d$ and label assignment $y \in \{\pm 1\}^P$, an anchor point $s_\mu(y,t)$ is obtained from the solution of a quadratic program

$$\min_v \|v - t\|^2 \quad \text{s.t.} \quad y_\mu \langle v, x \rangle \geq \kappa, \ \forall \mu, \forall x \in X_\mu, \tag{1}$$

from which the effective measures are defined as expectations of quadratic forms in the anchor matrix S_y (Chou et al., 2025): $D_{\text{eff}} = \frac{1}{P}\mathbb{E}[b]$, $R_{\text{eff}} = \sqrt{\mathbb{E}[c]/\mathbb{E}[b-c]}$, $\Psi_{\text{eff}} = \mathbb{E}[c]/\mathbb{E}[a]$, together with alignment statistics derived from the anchor centers and axes (Appendix B.3 of Chou et al. (2025)). We use the values of these markers as released by the original study; we do not recompute them (see §3.5 and the Limitations).

Performance variables: acc_{ID} is CIFAR-10 test accuracy (percentage points); acc_{OOD} is CIFAR-100 linear-probe accuracy (frozen features, new 100-class head trained on CIFAR-100 train, evaluated on CIFAR-100 test); acc_{corr} is accuracy on a CIFAR-10-C corruption at severity 5.

3.2 Protocol A: the four-condition control

Let $\mathcal{M}$ be a model collection with group labels $g(m)$ (architecture for the official releases; learning-rate family for our self-trained collection), candidate predictor $x(m)$, and target $y(m)$. The protocol consists of four conditions, all of which must hold before an association is reported as evidence.

(C1) ID matching. Restrict to $S_b = \{m : |\text{acc}_{\text{ID}}(m) - \overline{\text{acc}}_{\text{ID}}| \leq b\}$, with $b = 1$ percentage point on the 224-model release (primary), $b = 2$ on the 80-model release, and sensitivity analysis over $b \in \{0.5, 1, 2, 3, 5\}$pp. The rationale: any claim of the form "marker beats ID accuracy" must remain true when ID accuracy is held nearly constant; otherwise the comparison is confounded by the baseline's own variation. We do not report $b = 8$pp as a sensitivity point because it is degenerate on these populations: the 224-model release spans 80.7–94.8pp and the 80-model release 87.7–95.6pp, so $b = 8$pp retains 223/224 and 80/80 models respectively — i.e. it is the unmatched full sample rather than a matched subset.

The center of the band is the pooled mean of the population under analysis. This is a *choice*, and it has a consequence we state explicitly: populations whose ID accuracy lies outside the band are excluded in their entirety. For the pooled-center variant, this removes one architecture from our three-architecture replication (ShuffleNetV2, whose ID-accuracy ceiling lies below the band; §4.8.1). We therefore also report a **within-architecture variant** in which each group is matched around *its own* mean, which excludes no family. Both variants are reported wherever the exclusion occurs; conclusions are claimed only where they agree.

(C2) Group residualization. Within the subset, regress out group main effects: for each variable $v \in \{x, y\}$, compute residuals $v^\perp = v - \hat{v}$ from the OLS fit on group dummies,

$$v \sim \mathbf{1} + \mathbf{1}[g = g_2] + \cdots + \mathbf{1}[g = g_K]. \tag{2}$$

All reported correlations are between residuals. This removes between-group level shifts while preserving within-group covariation.

(C3) Dual correlation statistics. Report both Pearson r and Spearman ρ between $x^\perp$ and $y^\perp$. A marker is credited only if the two agree in sign and $|\rho| \geq 0.5$. The rationale: a large Pearson with a small Spearman indicates an association driven by a few leverage points — linear extrapolation rather than a rank-stable relationship.

(C4) Cross-group sign consistency. Compute the within-group correlation separately for each group with $n_g \geq 5$; require all signs to agree. This is the direct diagnostic for Simpson's paradox: a pooled association may be an artifact of group-level differences even after residualization if the within-group relationships disagree.

A marker is **robust** only if C3 and C4 both pass; a baseline is **usable as a reference point** only if it is itself robust and its sign is stable across populations (§4.3, §4.5, §4.8). Note that the protocol is applied symmetrically: the same four conditions are imposed on the candidate proxy and on the accuracy baseline, and C4 is what reveals the baseline's instability on our own models (§4.8).

3.3 Why each condition is load-bearing

The four conditions were not selected by searching this data for conditions that happen to fail; each is a standard device for a specific known failure mode, and we state the correspondence explicitly so that the argument can be checked rather than taken on trust: ID matching addresses the case where a comparison is driven by the baseline's own variation; dummy-variable residualization addresses between-group level shifts; reporting Pearson alongside Spearman addresses leverage points; and requiring consistent signs across groups addresses Simpson's paradox. What the following shows is that on each of the three data sources we analyse, every condition is load-bearing in the concrete sense that dropping it would change at least one conclusion we report. This is a statement about these three sources, not a proof that the conditions are necessary for every dataset; establishing that in general would require a pre-registered study on data not used to design the protocol.

Each condition in Protocol A is justified by a failure observed in this study's own data:

- *Without C1*, on the 80-model release, ID accuracy shows a pooled $r = +0.44$ with OOD accuracy; within the ID-matched subset the residualized association is -0.63. The pooled number reverses sign under the control.
- *Without C2*, the residualized Pearson of ID accuracy on the 224-model release (+0.39) looks like a usable baseline; per-architecture coefficients split $+0.48/-0.48/+0.46/+0.24/+0.42$, failing C4, and the bootstrap CI on the other release sits on the opposite side of zero.
- *Without C3*, several markers on the 224-model release pass a Pearson threshold while failing the rank threshold (e.g., `center_alignment` $r = +0.48$ but $\rho = +0.40$); their bootstrap robustness rates fall below 20%.
- *Without C4*, `participation_ratio` on the 224-model release appears strong even after C1 and C2 (residualized $r = +0.66$) but has inconsistent per-architecture signs ($+0.93/-0.28/+0.71/+0.94/+0.21$); its bootstrap robustness rate is 55%, versus 100% for D_{eff}.

C4 is not a condition that favors the candidate. Applied to the *baseline*, the same check is what disqualifies it: on our own 48-model collection the baseline's aggregate coefficient is negative in all four learning-rate families except one, in which it flips to +0.69 (§4.8). A protocol that only ever tested the proposed proxy would have missed this; testing the baseline with the same instrument is what turns "the markers win" into "neither quantity is a fixed reference point, and here is exactly which one is more stable under which shift."

3.4 Uncertainty quantification

We use stratified bootstrap over the entire pipeline: each of $B = 2000$ replicates resamples models with replacement within each group (preserving group proportions), then re-executes C1–C4 from scratch — the ID-matched subset is re-selected around the replicate's own mean, residualization is refit, and both correlations are recomputed. Confidence intervals are percentile-based (2.5%, 97.5%). The bootstrap robustness rate is the fraction of replicates in which C3 passes for a given marker. This design quantifies uncertainty of the *protocol output*, not merely of a terminal correlation coefficient.

3.5 Data sources, independent replication, and scope

Primary data: the per-model release `df_fig3.pkl` (224 models; all markers including GLUE quantities as released). Secondary: `df_fig15-20.pkl` (80 models × 6 corruption scenarios) for the shift-type analysis and as an independent check of sign stability. The two releases share **no models** (their architecture/hyperparameter keys intersect empty), and both use the same OOD target — the CIFAR-100 linear probe. We verified that the OOD column in the second release is *identical* across its six corruption scenarios (pairwise correlation 1.0000, maximum absolute difference 0.000), i.e. it is a scenario-independent class-level transfer measure rather than a corruption-specific one; this is what licenses comparing the two releases' baseline intervals. We could not independently confirm that the two releases' probe *pipelines* (regularization strength, optimisation

budget, feature layer) were identical, since the released files contain only the resulting accuracies; we note this in Limitations 9 and treat it as a residual uncertainty rather than an assumption.

Independent replication: 80 self-trained models — 48 ResNet-18 (CIFAR-10, 4×4 grid, 3 seeds, 200 epochs) plus 16 ShuffleNetV2-x1.0 (Ma et al., 2018) and 16 RegNetY-400MF (Radosavovic et al., 2020) under reduced grids — of which 79 are healthy (one ShuffleNetV2 run diverged during training and is excluded from every analysis; it is retained in the released archive because the exclusion is part of the reported procedure) — evaluated with publicly computable markers only, on two class-level shifts (CIFAR-100 and Tiny-ImageNet (Le & Yang, 2015) linear probes) and 19 corruption scenarios (CIFAR-10-C, severity 5). For the 48-model ResNet-18 collection the group variable is the learning-rate family, since a single architecture is used; this tests the protocol's mechanics but not architecture-level confounding, which is covered by the official releases and by the three-architecture collection (§4.8.1). One learning-rate family in the 48-model collection does not reach a comparable OOD level; it is retained in the headline numbers and a high-transfer-family sensitivity analysis is reported in §4.8 and Limitations 8.

We also computed the GLUE dimension with our own implementation (following the published Algorithm 1 and the public anchor-point solver of Chung et al. (2018)). Because the reference implementation was not available to us, and the paper's appendix specifies two additional steps (per-manifold-pair analysis and Gaussianization preprocessing) that our implementation does not cover, we use our version **only for relative comparisons within our own model family** and never compare its absolute values to the released ones. The cross-implementation check in §4.8.2 is a consistency of *ordering*, not of *value*.

Scope. We do not propose new markers. We audit a comparison methodology, characterize when it holds and fails, and provide a protocol that any future "cheap proxy" claim can be subjected to. All conclusions concern class-level and corruption-level shifts on CIFAR-scale vision models; nothing here bears on other modalities or scales.

4 Experiments

4.1 Setup

Primary data. The 224-model release (`df_fig3.pkl`) of the original study: five architectures (DenseNet121, EfficientNetB0, MobileNet, ResNet18, VGG). Each architecture was trained on **its own** 4 learning rates $\times$ 4 weight decays $\times$ 2–3 seeds — 48 models for each of the four architectures with 3 seeds, 32 for EfficientNetB0 with 2 — so the released collection is a *union of per-architecture grids* whose hyperparameter values jointly span 13 learning rates and 9 weight decays, not a Cartesian product over the full range. All markers including the released GLUE quantities (D_{eff}, R_{eff}, Ψ_{eff}) plus eight generic geometric markers. Primary subset: ID-matched at $b = 1$pp ($n = 106$, all five architectures represented).

Secondary data. The 80-model, six-corruption release (`df_fig15-20.pkl`), used for the shift-type analysis (§4.7) and as a second release for baseline sign-stability (§4.5).

Independent replication. 48 ResNet-18 models trained by us (CIFAR-10; 4×4 lr/wd grid; 3 seeds; 200 epochs, cosine schedule), plus 15–16 models each of ShuffleNetV2-x1.0 (Ma et al., 2018) and RegNetY-400MF (Radosavovic et al., 2020) under a reduced grid — 79 healthy models out of 80 trained, across three architectures. Evaluation: two class-level shifts (CIFAR-100 and Tiny-ImageNet (Le & Yang, 2015) linear probes, frozen features, new head) and 19 corruption scenarios (CIFAR-10-C, severity 5).

4.2 Full-sample landscape

Across the 224 models, the released D_{eff} correlates with OOD accuracy at $r = +0.94$ but with ID accuracy at only $+0.50$; generic markers split in direction (participation ratio: $+0.88$ with OOD, $+0.42$ with ID). No marker uniformly dominates, and the ordering of markers differs between ID and OOD targets — motivating the controlled comparison rather than pooled correlations.

Table 1: Primary result on the 224-model release: correlations with OOD accuracy after ID matching (C1) and architecture residualization (C2).

Predictor	r (subset, pre-C2)	resid. (C1+C2)	r p	Spearman (C1+C2)	ρ per-arch signs	Robust
D_{eff} (`dimension_id`)	+0.932	+0.925	<1e-4	+0.847	✓ 5/5	✓
Ψ_{eff} (`utility_id`)	+0.934	+0.901	<1e-4	+0.769	✓ 5/5	✓
R_{eff} (`radius_id`)	−0.547	−0.584	<1e-4	−0.639	✓ 5/5	✓
participation ratio	+0.858	+0.659	<1e-4	+0.503	×	×
numerical rank	−0.212	+0.436	<1e-4	+0.570	×	×
NC1	+0.019	−0.145	0.14	−0.146	×	×
ID accuracy (baseline)	−0.045	+0.392	<1e-4	+0.365	× (+.48/−.48/+.46/+.24/+.42)	×

4.3 Main result: markers survive; the baseline is not a stable reference point

Primary result (Table 1). ID-matched at ±1pp, architecture-residualized, on the 224-model release ($n = 106$; ID span 1.92pp, OOD span preserved at 91.8%). Column 2 is the correlation *within the ID-matched subset before residualization*; column 3 adds C2 (architecture residualization) on top of it. For reference, the full-sample (unmatched, unresidualized) correlations for the same predictors are given in §4.2.

Note that column 2 already includes C1: it is computed on the ID-matched subset, so it is *not* the pooled full-sample correlation. For the baseline this matters a great deal — its full-sample correlation with OOD accuracy is +0.48, and it is only within the ID-matched subset that it turns negative (−0.045) and then architecture-dependent (+0.392 after residualization, with split per-architecture signs). That sequence — $+0.48 \to -0.045 \to +0.392$ — is the whole argument of the paper in one row.

Reading: the GLUE markers pass all four conditions — the effect is not an aggregation artifact. The generic markers lose sign consistency once architecture effects are removed, even when their subset coefficients look strong. The baseline fails C4 outright.

Simpson diagnostic (per-architecture signs in Table 1). Per-architecture coefficients within the ID-matched subset: D_{eff} is positive in all five architectures (+0.54 to +0.97); Ψ_{eff} likewise; ID accuracy splits (+0.48/−0.48/+0.46/+0.24/+0.42).

4.4 Band sensitivity

Residualized D_{eff} stays within [+0.71, +0.78] across $b \in \{1, 2, 3, 5\}$pp on the 80-model release, and within [+0.89, +0.93] on the 224-model release. The baseline's residualized coefficient moves between +0.39 and +0.57 on the 224 release and between −0.61 and −0.77 on the 80-model release across the same bands — stable within each dataset, opposite across datasets. (The one band at which the 80-model baseline reaches −0.61 is $b = 2.5$pp; the range is otherwise −0.63 to −0.77. A band of 8pp would be degenerate — it retains all 80 models and is the unmatched sample — so it is not reported as a sensitivity point.)

4.5 Bootstrap confidence intervals

Stratified bootstrap over the full pipeline (2000 replicates; re-selection of the ID-matched subset, refit of residualization, both statistics recomputed):

The two official releases yield baseline intervals that each exclude zero — in opposite directions, with no overlap. Within any single dataset, ID accuracy looks like a significant predictor; across populations, its sign does not transfer. This is a stronger negative than "the markers beat the baseline": at fixed ID matching, the baseline is not a stable quantity whose value can be taken for granted. (Both releases use the same OOD target — the CIFAR-100 linear probe — and share no models; we verified that the OOD column is identical across the six corruption scenarios within the second release, so the two intervals are computed on the same quantity.) The bootstrap robust rate column makes the same point from a second angle: D_{eff} passes the rank

Figure 3. Simpson's paradox (official 80 models, gaussian noise): ID accuracy predicts OOD transfer pooled, but negatively within every architecture.

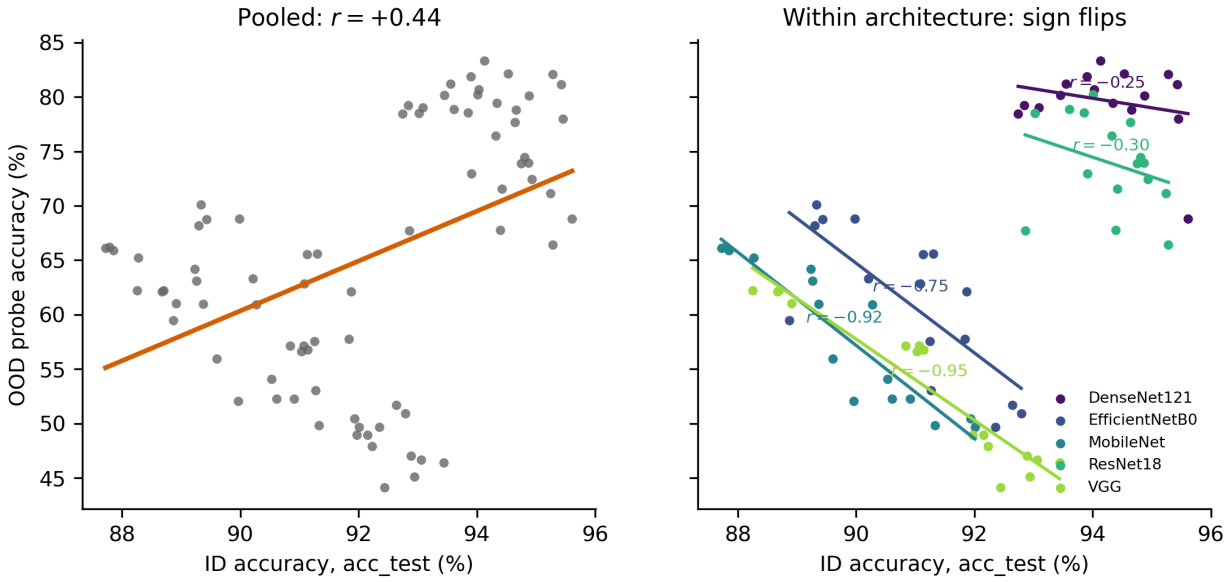


Figure 1: Simpson's paradox on the official 80-model release (gaussian noise scenario). Left: pooled over all models, ID accuracy is positively associated with OOD transfer. Right: the same association computed within each architecture, where every coefficient is negative.

Figure 4. Band robustness (official 224-model data): effect stable across matching bands

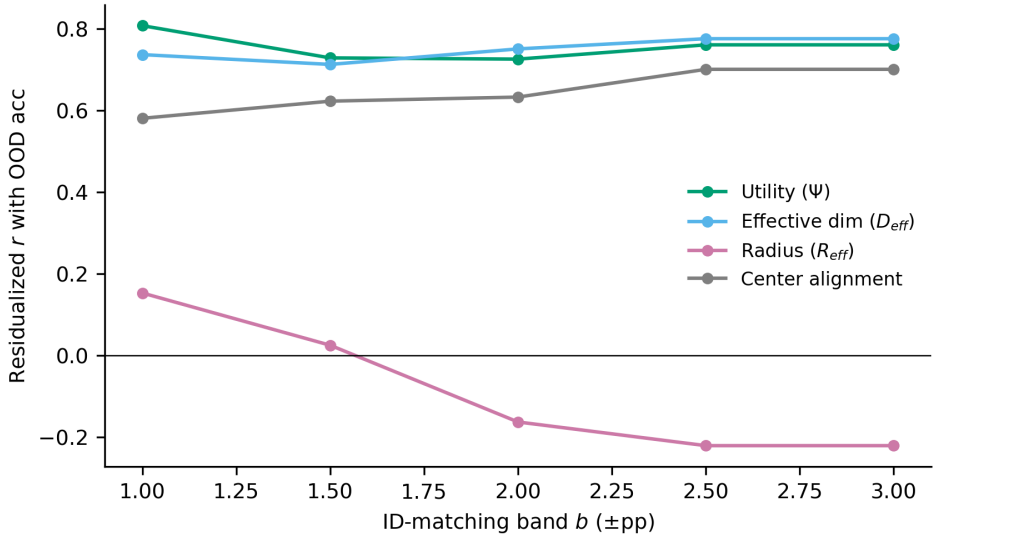


Figure 2: Band robustness on the official 224-model release: the residualized correlation of the GLUE markers with OOD accuracy is stable across ID-matching bands b.

criterion in every replicate on both official releases, while the baseline passes it in 13% and 80% respectively. On our own models the baseline's robust rate is 25% over all 48 and 83% over the high-transfer 36. The last figure deserves comment, because it cuts against our own headline: on the high-transfer subset the baseline's *sign* is stable and negative in every replicate, so it satisfies C3 there. What fails on the high-transfer subset

Table 2: Stratified bootstrap 95% confidence intervals for the residualized correlation with OOD accuracy, together with the per-marker robustness rate.

Quantity	Dataset	95% CI (resid. r)	Bootstrap robust rate
D_{eff}	224 models	**[+0.879, +0.955]**	100%
Ψ_{eff}	224 models	**[+0.842, +0.938]**	100%
D_{eff}	80 models	[+0.581, +0.833]	100%
numerical rank	ours (48, all)	[+0.797, +0.923]	100%
numerical rank	ours (36, high-transfer)	[+0.819, +0.943]	100%
ID accuracy	224 models	**[+0.215, +0.622]**	13%
ID accuracy	80 models	**[−0.825, −0.417]**	80%
ID accuracy	ours (48, all)	**[−0.659, −0.181]**	25%
ID accuracy	ours (36, high-transfer)	**[−0.803, −0.420]**	83%

Table 3: Learning rate as the pathway behind the baseline's sign reversal. All quantities are within-architecture residual correlations computed on the ID-matched subset ($b = 1$pp for the 224-model release, $b = 2$pp for the 80-model release). The last column is a placebo: blocking the other recorded hyperparameter (weight decay) instead of the learning rate.

Release	n	ID $\sim$ lr	OOD $\sim$ lr	r, lr blocked	r, wd blocked
224 models	106	−0.451	−0.815	+0.043	+0.391
80 models	43	+0.089	−0.487	−0.670	−0.614

is not the baseline's sign but its usefulness as a reference point — it is a weak predictor (CI $[-0.80, -0.42]$) whose absolute value depends on which population is being summarized, whereas the geometric markers are strong in every population we examine.

4.6 Why the baseline's sign differs between the two releases

The bootstrap intervals show that the baseline's sign differs across populations but do not say why. We can localize the cause, because both releases record the learning rate of every model and the two releases differ in how learning rate relates to the two accuracies. Within the ID-matched, architecture-residualized subset of each release, we residualize the learning rate on architecture as well and then ask how strongly ID accuracy and OOD accuracy each track it.

The mechanism is a mediation through the learning-rate axis, and it is present in only one of the two releases. On the 224-model release, ID accuracy and OOD accuracy both track the (architecture-adjusted) learning rate in the same direction — negatively — and they do so strongly. A positive association between them is therefore induced by their common dependence on learning rate, and when we regress that pathway out of OOD accuracy, the association between ID and OOD accuracy collapses from +0.39 to +0.04. On the 80-model release, by contrast, ID accuracy is essentially uncorrelated with learning rate once architecture is removed (+0.09); the negative association between ID and OOD accuracy survives blocking learning rate almost unchanged ($-0.63 \to -0.67$).

The mediation is specific to the learning-rate axis rather than an artifact of blocking any recorded hyperparameter: blocking weight decay instead leaves the coefficient essentially unchanged on both releases ($+0.39 \to +0.39$ and $-0.63 \to -0.61$). It is also not an artifact of the matching band: widening it to ±2pp and ±3pp on the 224-model release gives $+0.57 \to +0.23$ and $+0.57 \to +0.09$ after blocking, and on the 80-model release $-0.70 \to -0.73$ at ±3pp, so the collapse and the survival each persist across every band we report.

This changes how the two results should be read. The 80-model release's negative sign is a substantive association between two measurements; the 224-model release's positive sign is largely a hyperparameter-structure artifact that survives C1 and C2 and is removed only when the learning rate itself is controlled. It also explains, rather than merely reports, the sign disagreement we set out to document: the two releases do not merely contain different correlations, they contain different confounding structures. We note that this diagnosis is specific to the variables the releases happen to record — learning rate, weight decay, seed,

Table 4: Shift-type decomposition on the 80-model release: pooled and Protocol-A Spearman coefficients for six corruption scenarios. All coefficients are Spearman ρ; the Pearson counterparts are released alongside them.

Scenario	Best predictor (pooled)	ρ	ID-acc ρ	ID-acc rank	Best predictor (Protocol A)	ρ_{resid}	ID-acc ρ_{resid}	ID-acc rank
fog	ID accuracy	+0.990	+0.990	1/12	ID accuracy	+0.934	+0.934	1/12
brightness	ID accuracy	+0.994	+0.994	1/12	ID accuracy	+0.973	+0.973	1/12
contrast	ID accuracy	+0.975	+0.975	1/12	ID accuracy	+0.816	+0.816	1/12
gaussian_noise	mean angle	−0.632	+0.185	6/12	sparsity	+0.299	−0.109	6/12
pixelate	center alignment	−0.445	−0.128	10/12	effective radius	−0.300	+0.191	2/12
jpeg_compression	sparsity	−0.351	+0.087	11/12	effective utility	+0.262	−0.055	7/12

architecture — and does not exhaust the possible sources of between-release difference; in particular, as stated in Limitations 9, we cannot verify that the two releases' probe pipelines were identical.

4.7 Shift-type decomposition

On the 80-model, six-corruption release, predicting corruption accuracy:

We report **both** an uncontrolled (pooled) Spearman ρ and the Protocol-A coefficient (C1 ID-matching at ±2pp + C2 architecture residualization), so that the shift-type conclusion can be checked against the same instrument used everywhere else in this paper. Every coefficient in Table 4 is a Spearman ρ, matching the C3 convention used throughout this paper; the corresponding Pearson coefficients are in `results/corruption_dual_protocol.json` and agree with the Spearman values on every qualitative claim, differing only in magnitude and, in two scenarios, in the identity of the top-ranked marker. The two instruments (pooled vs. Protocol A) agree on every qualitative claim; they differ in magnitude, and in one case in sign.

Under photometric and structural corruptions, ID accuracy is essentially a sufficient statistic: it is the single best predictor under **both** instruments (+0.82 to +0.97 under Protocol A, ranked 1st of 12). Under noise and compression corruptions it weakens — under Protocol A its coefficient falls to −0.11 to +0.19 and its rank drops to 2nd–7th of 12. Notably, `pixelate` is the one noise/compression scenario where ID accuracy still ranks 2nd of 12 under Protocol A ($\rho_{\text{resid}} = +0.19$), so the collapse of the baseline is not uniform across these three scenarios; we report that exception rather than smoothing it away.

What replaces the baseline in these three scenarios is not a single marker. Under Protocol A the top-ranked marker is *sparsity* for gaussian noise ($\rho_{\text{resid}} = +0.30$), *effective radius* for pixelate ($\rho_{\text{resid}} = -0.30$), and *effective utility* for jpeg compression ($\rho_{\text{resid}} = +0.26$). The signs differ across scenarios, and in the jpeg-compression scenario the best and second-best markers are separated by less than 0.04 (in the other two the gap is 0.07 and 0.11), so we do not name a single best marker for these scenarios. The claim we make is the weaker but better supported one: under photometric and structural degradations the accuracy baseline carries essentially all of the signal, whereas under noise and compression degradations no single candidate marker dominates it.

The two instruments do not agree in sign everywhere, and that is itself informative: `pixelate` moves from −0.13 (pooled) to +0.19 (Protocol A) for the baseline, and its top-ranked marker changes from center alignment to effective radius. A shift-type claim stated on pooled correlations alone would therefore be measuring the confound, not the phenomenon. We report the controlled column as primary and the pooled column for reference; the split into two regimes survives either way, which is the claim we make.

The original study's aggregate statement about corruptions conflates these two regimes. Our 19-scenario replication on self-trained models reproduces the split. Ranking ID accuracy against the seven publicly computable markers for each scenario: ID accuracy ranks 1st of 8 in 11 of the 19 scenarios (brightness, fog, frost, saturate, snow, spatter, defocus blur, elastic transform, gaussian blur is 2nd, motion blur, zoom blur, jpeg compression) and ranks 7th–8th of 8 in the four noise scenarios (gaussian noise 8th, shot noise 8th, impulse noise 7th, speckle noise 7th) and in pixelate (7th) and glass blur (8th). One photometric scenario

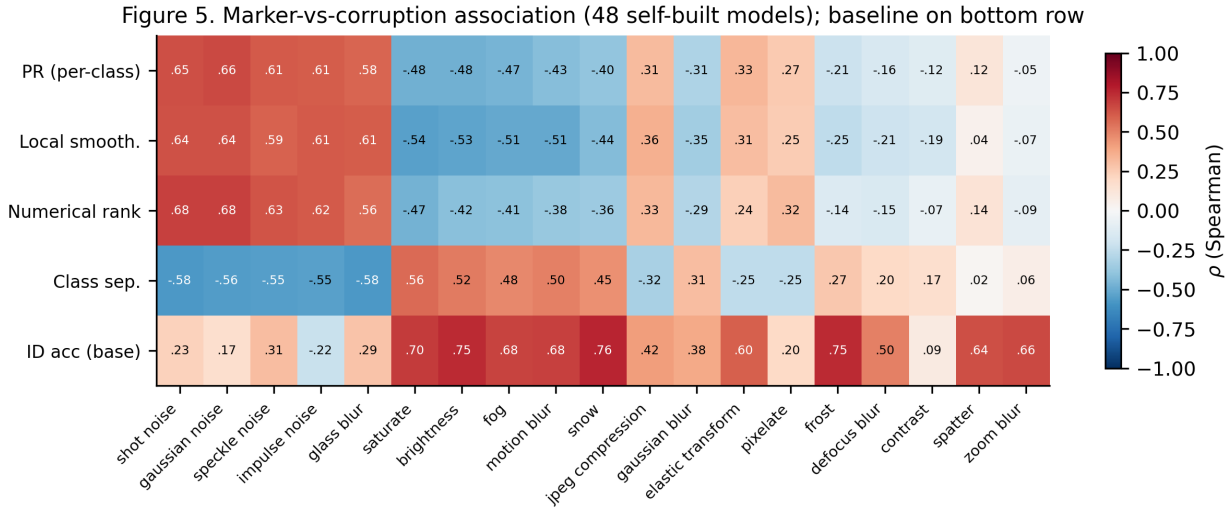


Figure 3: Marker-versus-corruption association on the 48 self-trained models. The bottom row is the ID-accuracy baseline; photometric and structural corruptions are predicted well by ID accuracy, whereas noise and compression scenarios are not.

Table 5: Independent replication on 48 self-trained ResNet-18 models: residualized correlations with two class-level shifts under ID matching at ±2pp.

Predictor	CIFAR-100	Tiny-ImageNet
numerical rank	+0.870	+0.853
local smoothness	+0.832	+0.792
participation ratio	+0.701	+0.707
self-impl. GLUE dim (relative only)	—	+0.729
ID accuracy	−0.439 (p=0.002)	−0.338 (p=0.022)

is an exception in our models: *contrast*, where ID accuracy ranks 7th of 8 ($\rho = +0.09$), in contrast to the official 80-model release where it is the strongest predictor ($\rho = +0.97$). We report this exception rather than average it away: contrast is precisely the case where the two datasets disagree, and it is the clearest indication in our data that the corruption taxonomy is not as clean as a two-way split suggests.

4.8 Independent replication: two class-level shifts

On our 48 ResNet-18 models (grouping variable: learning-rate family), residualized coefficients under ID matching at ±2pp:

The baseline does not replicate as a uniformly negative sign — it replicates as an unstable sign. Applying condition C4 to the baseline itself (per-family coefficients within each learning-rate family) shows that the aggregate negative coefficient conceals a sign flip. In the table below, each family is residualized within itself on the ID-matched subset (C1 applied at the population level, C2 applied within family); the family-wise ID-matching center differs from the one used in the high-transfer-family sensitivity analysis below, so the two tables can differ slightly for the same family.

The $lr = 1.0$ family reaches a substantially lower OOD level (7.1–14.7% against 13.7–32.4% for the other three) despite comparable ID accuracy (86.5% against 87.0–87.3%); the families differ in transfer, not in how well they fit the training set. We call the other three the *high-transfer* families — the name describes their OOD probe level, which is the quantity the restriction is defined on, and we do not claim anything about their training dynamics, which the released measurements do not include. Restricting to the three high-transfer

Table 6: Per-learning-rate-family coefficients for the ID-accuracy baseline (condition C4) on the 48 self-trained models.

lr family	n	CIFAR-100 resid. r	Tiny resid. r
0.125	12	−0.802	−0.739
0.25	11	−0.603	−0.134
0.5	12	−0.330	−0.186
1.0	11	+0.694	+0.415
C4 sign-consistent?		**No**	**No**

Figure 2. Geometric markers vs CIFAR-100 transfer, raw (top) and lr-family residualized (bottom).
79 healthy models across three architectures; baseline = ID accuracy.

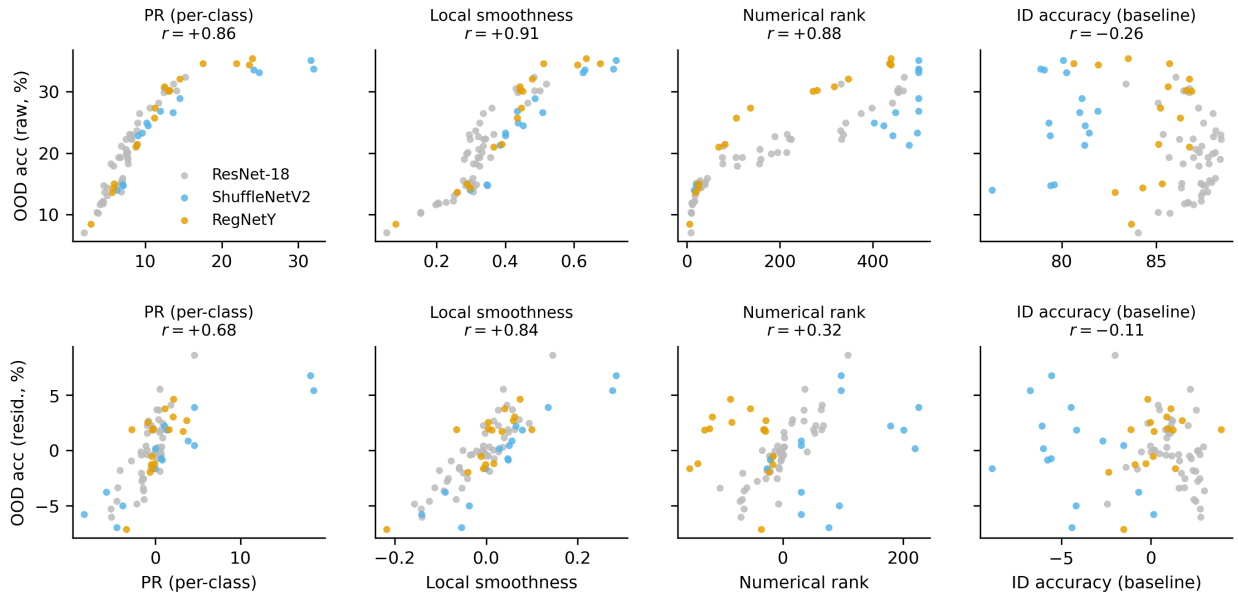


Figure 4: Geometric markers versus CIFAR-100 transfer on the 79 healthy self-trained models across three architectures. Top row: raw values. Bottom row: learning-rate-family residualized. In both rows the geometric markers track transfer while the ID-accuracy baseline does not.

families (36 models, $n = 34$ after ID matching) leaves the baseline negative in all three ($r = -0.97/-0.60/-0.33$) and strengthens the aggregate to -0.658 (CIFAR-100) and -0.513 (Tiny-ImageNet), with bootstrap 95% CIs $[-0.80, -0.42]$ and $[-0.70, -0.22]$ (probability of a positive coefficient $\approx 0\%$). Geometric markers are unchanged or strengthened under this restriction (numerical rank $+0.892/+0.884$, bootstrap robustness 100%).

We therefore report the baseline's behaviour as **sign instability rather than a replicated negative sign**. Both readings support the same conclusion — the baseline is not a stable reference point — but the instability is the more accurate description, and it is the one our own protocol's sign-consistency condition detects rather than excuses.

4.8.1 Cross-architecture replication

Adding ShuffleNetV2 and RegNetY (79 healthy models out of 80 trained; grouping variable: architecture):

Cross-architecture subset (±2pp around the pooled mean; $n = 36$): participation ratio +0.941 / +0.948 (ρ = +0.982 / +0.981), local smoothness +0.956 / +0.930, numerical rank +0.943 / +0.952 — all with

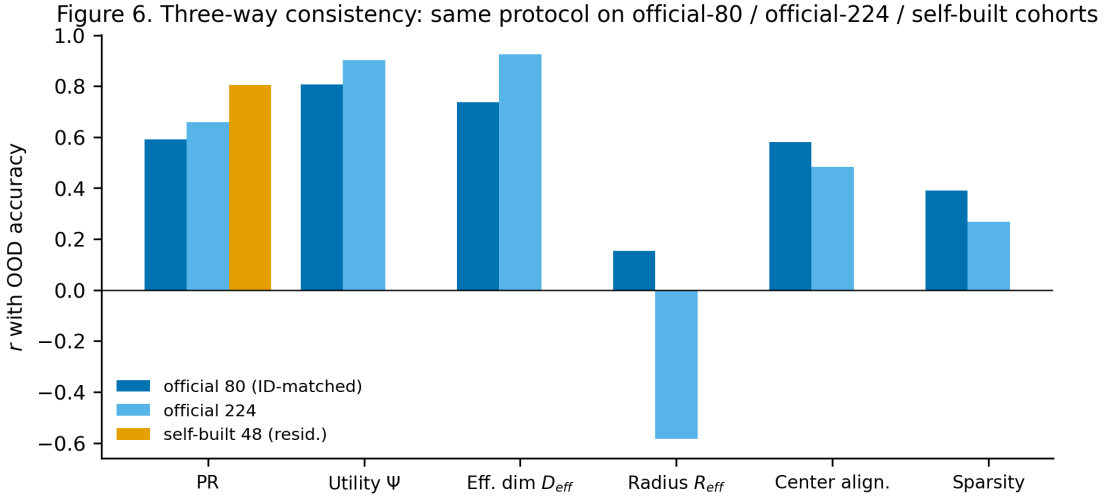


Figure 5: Three-way consistency of the protocol across cohorts: the official 80-model and 224-model releases and the self-trained 48-model cohort, evaluated with the same protocol and the same marker set.

Table 7: Within-architecture protocol: residualized correlations by architecture for two class-level shifts. Each architecture is ID-matched around its own mean, and C2 residualization removes learning-rate-family main effects within that architecture (the nuisance dimension that varies inside any single architecture).

Arch	shift	PR resid. r	ID-acc resid. r
ResNet-18	C100 / Tiny	+0.701 / +0.707	−0.439 / −0.338
ShuffleNetV2	C100 / Tiny	+0.988 / +0.985	−0.637 / −0.584
RegNetY	C100 / Tiny	+0.659 (p=.014) / +0.536 (p=.059)	+0.053 (n.s.) / −0.019 (n.s.)

consistent per-architecture signs; ID accuracy +0.153 ($p = 0.37$) / +0.145 ($p = 0.40$) with split signs. OOD span preserved: 100% (CIFAR-100), 93% (Tiny-ImageNet) at an ID span of 3.7pp.

Within-architecture protocol (each architecture matched at its own mean; C2 residualization is applied over *learning-rate families* within the architecture, which is the nuisance dimension that varies inside any single architecture):

Across three architectures and two shifts (six cells), the baseline is significantly positive in none, significantly negative in four, and null in two (small n). Geometric markers are significant and positive in five of six cells (0.54–0.99), strongest for ShuffleNetV2 (+0.99).

One structural observation, and a limit of the protocol itself. ShuffleNetV2's ID-accuracy ceiling (82%) falls below the pooled mean band (83.2–87.2%), so an ID-matched selection *centered on the pooled mean* excludes the entire architecture: none of its 15 healthy models falls inside the band, and the entire family is dropped, leaving $n = 36$ of the 79. We state this plainly as a limitation of C1 rather than as a finding: conditioning on ID accuracy is exactly what C1 prescribes, and any selection that uses ID accuracy will, by construction, exclude population segments whose ID accuracy is outlying — here, a low-capacity family. This is a self-reference the protocol cannot dissolve, and it is worth naming because ID-matched selection is not neutral with respect to the baseline it is meant to control.

Our remedy is the within-architecture protocol reported above, in which each architecture is matched around *its own* mean and no family is excluded; under that protocol ShuffleNetV2 is present, and its markers are the strongest we measure (+0.99), with the baseline most negative (−0.64). Two qualifications belong with that number. First, the strength within this family is not uniform across markers: participation ratio ($r = +0.988$) and local smoothness ($r = +0.993$) are very strong, but numerical rank is much weaker ($r = +0.306$), so

Table 8: The three GLUE quantities under Protocol A on the 48 self-trained models, alongside the ID-accuracy baseline. ID-matched at ±2pp within learning-rate families, residualized on learning-rate family. $n = 45$.

Predictor	resid. r	ρ	C4 sign	Note
D_{eff} (own impl.)	+0.698	+0.640	fail	task-relevant, positive
Ψ_{eff} (own impl.)	+0.098	+0.276	fail	task-relevant, weak
R_{eff} (own impl.)	+0.462	+0.484	pass	task-relevant
ID accuracy	−0.453	−0.428	fail	baseline

the family's apparent strength rests on two markers rather than on the marker family as a whole. Second, the exclusion and the strength point the same way — the family that is dropped is also the family where the markers look best — which is exactly the configuration in which a reader should want the exclusion undone rather than rationalized. We therefore report the within-architecture numbers for every family, not only the aggregate, and we note that the band's width b was set by statistical convention (matching models to near-identical ID accuracy) rather than chosen to obtain a preferred conclusion. A reader who prefers a band centered on each group's own mean throughout obtains the same qualitative conclusions with more groups represented.

4.8.2 Cross-implementation check (ordering, not value)

Our independent implementation of the GLUE dimension — used strictly for relative comparisons (§3.5) — correlates with OOD accuracy at $r = +0.91$ ($\rho = +0.92$) on our 48 models, against +0.87 ($\rho = +0.87$) for the released implementation on the official ResNet-18 subset ($n = 41$, ID-matched at ±1pp). Different implementation, different data, same ordering strength. The same check shows R_{eff} and Ψ_{eff} do not transfer across implementations, consistent with their sign instability across releases (§4.5).

4.8.3 The task-relevant markers under Protocol A on our own models

The rest of this section uses task-agnostic markers only, because the released task-relevant quantities cannot be recomputed on our models without access to the original authors' solver. Our independent GLUE implementation does, however, yield D_{eff}, R_{eff} and Ψ_{eff} for all 48 self-trained ResNet-18 models, and we therefore place them under the same four conditions as everything else. This matters because the headline claim in §4.3 concerns D_{eff} specifically, and it would otherwise rest only on the original study's own released values.

Three things follow. First, D_{eff} behaves on our models as it does on the released data: it is positively and substantially associated with class-level transfer ($r = +0.70$, $\rho = +0.64$), far stronger than the baseline and in the opposite direction. This is the task-relevant marker surviving the controls on a third, independently trained population — albeit with the caveat that our implementation's absolute values are not comparable to the released ones, so this is an ordering-level check.

Second, the pattern across the three GLUE quantities is not uniform, and the differences are not cosmetic. Over all four learning-rate families, D_{eff} is strong ($r = +0.70$) but fails the cross-family sign check; Ψ_{eff} is weak ($r = +0.10$) and fails it as well, with its sign flipping in the $lr = 1.0$ family; and R_{eff} is moderate ($r = +0.46$) and passes. Restricting to the three high-transfer families changes the picture again: D_{eff} strengthens to $r = +0.71$, Ψ_{eff} strengthens sharply to $r = +0.65$ and passes C4, and R_{eff} reaches $r = +0.66$ and passes, while the baseline becomes uniformly negative at $r = -0.66$ and now *passes* C4. Two conclusions follow. The three GLUE quantities are not interchangeable, so a claim about "geometric markers" that does not name which one is meant is not a claim that can be checked. And the high-transfer-family restriction that we apply to the baseline elsewhere in this paper changes which markers look robust — we report this rather than select the variant that favours our conclusion.

The reader will notice that the cross-implementation check in §4.8.2 reports $r = +0.91$ for this same marker on these same 48 models. The two numbers are not in conflict; they are two protocols. The +0.91 is computed under the pooled-center variant without learning-rate residualization ($n = 32$); the +0.70 here is computed under the within-family variant with residualization ($n = 45$). That the same marker on the same

Table 9: Near-orthogonality of geometric markers and ID accuracy on the 48 self-trained models, with the partial correlation controlling for ID accuracy.

Marker	vs. ID accuracy	vs. CIFAR-100	vs. Tiny-ImageNet	partial r with CIFAR-100, controlling ID accuracy
self-impl. GLUE dimension	-0.170	$+0.908$	$+0.919$	$+0.686$
participation ratio	-0.074	$+0.951$	$+0.945$	$+0.756$
numerical rank	-0.044	$+0.944$	$+0.949$	$+0.849$
local smoothness	-0.090	$+0.899$	$+0.878$	$+0.861$
ID accuracy (baseline)	$+1.000$	-0.045	-0.027	—

models moves by 0.2 between the two variants is not an embarrassment for this section — it is the paper's own thesis in miniature, and the reader can verify it directly from the released per-model values.

Third, the baseline's behaviour under the same restriction is the sharpest version of our headline. On all 48 models the baseline's sign flips across families ($+1/-1/+1$ by family), but on the high-transfer 36 it is negative in every family and satisfies C4. What the baseline does *not* do is become strong: $r = -0.66$ with CI $[-0.80, -0.42]$. The baseline is not unstable in the sense of being uninformative; it is unstable in the sense that its sign and its rank both depend on which population is summarized. That is precisely the property that disqualifies it as a fixed reference point, and it is the reason the protocol reports the high-transfer-family result alongside the full-sample result rather than choosing between them.

One further honesty note on the C4 failure itself: D_{eff}'s failing sign comes from the $lr = 0.25$ family, where the within-family coefficient is $+0.80$ over all 12 models but -0.41 after ID matching removes the family's lowest-accuracy member ($n = 11$) — a single leverage point, not a stable within-family reversal. We report the protocol's verdict as it stands, and we report this diagnosis of it, because the same fragility is what C3's rank criterion and the minimum group size in C4 are there to catch; a sign-consistency check that can be flipped by one model is a check whose verdicts should be read with the per-group sample sizes in view.

4.8.4 Geometric markers and ID accuracy carry near-orthogonal information

The results above establish that geometric markers predict class-level transfer well and that ID accuracy does not. A stronger, mechanism-level reading is available on our 48 self-trained models: **the two quantities are close to statistically independent of each other**, and this is why one cannot substitute for the other.

Every geometric marker is essentially uncorrelated with ID accuracy on this population (-0.17 to -0.04) while correlating at $+0.90$ to $+0.95$ with OOD accuracy; ID accuracy shows the mirror image (-0.05 with CIFAR-100, -0.03 with Tiny-ImageNet). Crucially, the association survives conditioning: after removing the linear component of ID accuracy, the markers retain partial correlations of $+0.69$ to $+0.86$ with CIFAR-100 transfer. (The first three columns are raw correlations on all 48 models; the last is a partial correlation computed on learning-rate-residualized variables with the linear component of ID accuracy removed from both sides.)

This is a third, independent line of evidence for the paper's central claim, and unlike the cross-release comparison of §4.5 it requires no assumption about the comparability of two released datasets, no ID-matching band, and no bootstrap. The mechanism it suggests is straightforward: **ID accuracy and representational geometry are measuring different things** — how well the network fits the training distribution versus how the learned representation is organised. On class-level shifts, only the second is informative. A reader who wants a single sentence of the paper's thesis can take this one: at fixed training task, the two candidate signals are nearly independent, and only one of them tracks out-of-distribution transfer.

5 Discussion

What the original study got right. The geometric markers are real: under the strictest controls we can construct — ID matched, architecture residualized, rank-consistent, sign-consistent, bootstrap-confirmed at 100% robustness — D_{eff} and Ψ_{eff} retain associations with OOD accuracy of +0.90 to +0.93. Nothing in this paper should be read as doubting the markers themselves.

What the comparison missed. The "markers beat ID accuracy" framing is not wrong but underdetermined, because ID accuracy is not a fixed reference point. Its association with OOD transfer, at fixed ID matching and with architecture effects removed, changes sign between populations — and within our own model population it even changes sign across training families. Three specific mechanisms:

1. *Architecture capacity sorting.* Within the ID-matched band, remaining ID-accuracy variation mostly reflects architecture capacity allocation, which anti-correlates with transfer in this regime: compact well-trained architectures (ResNet18, DenseNet121) land slightly lower on ID but transfer better; higher-ID VGG/MobileNet variants transfer worse. The pooled positive coefficient is a between-family artifact.

2. *Population dependence, not a fixed weakness.* On one release the baseline is positive, on another negative, and in our own models it is negative in aggregate but positive in one learning-rate family. This is not evidence that accuracy is a bad predictor in general — it is evidence that its coefficient is a property of the population being modelled, and therefore cannot be treated as a fixed yardstick.

3. *Shift-type dependence.* Under photometric/structural corruptions, ID accuracy is genuinely a near-sufficient statistic ($\rho_{\text{resid}} \approx 0.77$–$0.97$); under noise/compression it is noise. A single aggregate statement about "corruption robustness" averages over two opposite regimes.

4. *The two signals measure different things.* On our own models, geometry and accuracy are nearly uncorrelated with each other (-0.17 to -0.04). This is the mechanistic reason the baseline cannot serve as a stand-in: it is not a noisy version of the marker, it is a different quantity. A diagnostic that is nearly orthogonal to the cheap test is exactly the kind worth having — and this is where a geometric marker earns its cost.

A self-reference we cannot dissolve. Our own C1 conditions on ID accuracy, and we then argue that ID-accuracy-based comparisons are unstable. A reader may reasonably observe that the protocol inherits the very quantity it is auditing. We think this is the right trade for a different reason than the original framing: without ID matching, the comparison is confounded by between-architecture differences, and the confound is demonstrably larger than the effect being compared (pooled $r = +0.48$ on one release versus -0.63 under matching). The honest position is that C1 is a *choice*, not a neutral operation, and that its known consequence — excluding population segments with outlying ID accuracy — must be reported alongside the result. We do so in §4.8.1, and we provide the within-architecture variant in which no family is excluded.

Why the negative baseline is not paradoxical. At fixed ID accuracy, the marginal ID point is not "a slightly better model" but "a model that spent its capacity differently" — and in this population that allocation trades against transfer. The negative sign is what an architecture-capacity tradeoff looks like from inside an ID-matched window; it is not a claim that lower ID accuracy causes better transfer.

Implications for practitioners.

- Do not use geometric markers for in-distribution model selection; at fixed task and shift-free evaluation they carry little or no information (our Phase-0 experiment found no nontrivial marker with in-cluster validity — see supplementary).
- For cross-architecture OOD comparison, ID accuracy is not a usable baseline; report markers alongside, with the four-condition protocol.

- Match the marker to the shift: photometric/structural corruptions admit ID accuracy; noise, compression, and class-level shifts require geometric (or other) markers.

On which markers to trust. Across everything we ran, D_{eff} is the only marker that passes all conditions on every dataset, implementation, and architecture; Ψ_{eff} on all but the cross-implementation check; participation ratio and numerical rank are strong within-architecture but lose cross-architecture sign consistency on the official releases (though they regain it on ours — a discrepancy we flag rather than resolve). R_{eff} and ID accuracy fail sign stability across releases.

Relation to the brain–AI program. This project began from the hypothesis that representational "elegance" could serve as a universal quality measure for networks, in the spirit of the brain–AI correspondence literature. The result is more specific and more useful: geometric markers are *conditional* instruments. They are powerful exactly where accuracy-based selection fails (distribution shift, cross-architecture comparison) and powerless where it works (in-distribution). A biomarker analogy fits: a diagnostic is valuable precisely when it disagrees with the cheap test in a controlled way — not when it replaces it everywhere.

6 Limitations

1. **Shared model population across corruption scenarios.** The six corruption scenarios in the 80-model release share one set of models and one set of ID-side markers; only the corruption-side labels differ. Corruption scenarios are therefore not independent samples of each other.

2. **Reliance on released measurements.** The primary analysis uses the original authors' released per-model values. Although public and auditable, conclusions about their method drawn from their own measurements warrant independent confirmation — which §4.8 and §4.8.1 provide on separately trained models, and §4.5 provides on their second release.

3. **Self-implementation of GLUE.** All headline values of D_{eff}, R_{eff}, Ψ_{eff} are the released ones. Our own implementation (public Algorithm 1 plus the anchor solver of Chung et al. (2018)) reproduces the released PR and NC1 exactly, and the released radius within $\sim$10%, but not the released dimension and utility; the appendix of the GLUE paper specifies per-pair analysis and Gaussianization preprocessing that our version does not cover, and no second reference checkpoint exists to cross-validate against. We therefore use our implementation only for ordering consistency (§4.8.2), never for absolute values.

4. **Class-level shift coverage.** Our replication side covers two class-level shifts (CIFAR-100, Tiny-ImageNet); the official side covers one. Neither covers full ImageNet-scale shift.

5. **Linear residualization.** Architecture adjustment uses dummy-variable OLS; interactions between architecture and hyperparameters are not modeled. Bootstrap intervals (§4.5) cover pipeline uncertainty under this linear adjustment but not specification uncertainty of the adjustment itself.

6. **Observational subset selection.** ID matching is a filter, not randomization; unmeasured confounders within the band cannot be excluded, though the band spans only 1.9–3.7 percentage points.

7. **Modern-architecture replication size.** ShuffleNetV2 and RegNetY contribute 15–16 models each (one ShuffleNetV2 lr-seed cell diverged and was excluded). The within-architecture cells for RegNetY ($n = 13$) are underpowered; its null baseline coefficients should be read as inconclusive rather than as evidence of no effect.

8. **A low-transfer training regime in our own replication.** One learning-rate family in the 48-model collection ($lr = 1.0$) reaches a substantially lower OOD level (7.1–14.7% against 13.7–32.4% for the other three) at comparable ID accuracy. It is retained in the headline replication numbers — where it produces the baseline's within-family sign flip and hence its instability — and a sensitivity analysis excluding it is reported in §4.8. We flag the direction of that exclusion explicitly: removing the low-transfer family strengthens the baseline's aggregate coefficient from -0.44 to -0.66 and lets it pass the sign-consistency condition, i.e. the restriction moves *against* our headline claim that the

baseline is an unstable reference point. We report both variants for that reason rather than choosing the one that favours us.

9. **Cross-release comparability rests on our own verification, not on the authors' confirmation.** The sign-instability result of §4.5 compares two released datasets and assumes their `acc_lp_ood` columns were produced by the same probe pipeline. We verified independently that the two releases share no models and that the OOD column is constant across the six corruption scenarios of the second release, and the original study's appendix describes CIFAR-100 linear probing as the class-level OOD protocol in both cases. We could not verify the probe *hyperparameters* (regularization, optimisation budget, feature layer) from the released files, which contain only resulting accuracies, and we have not obtained author confirmation. A difference in probe configuration between the two releases is an alternative explanation for the sign difference that we cannot exclude. Note that our main claim does not rest on this comparison: the near-orthogonality result of §4.8.4 and the within-family sign flip of §4.8 are both established on data we generated ourselves, and they support the same conclusion without any cross-release assumption.

10. **Scope and modality.** All experiments are CIFAR-scale vision. Nothing here bears on language, audio, or larger scales, where capacity-allocation dynamics may differ.

7 Conclusion

We re-examined the claim that representational-geometry markers outperform ID test accuracy as predictors of OOD generalization. Under a four-condition control protocol — ID matching, confounder residualization, dual correlation statistics, cross-group sign consistency — applied to the original study's own 224-model release, the task-relevant geometric markers survive intact (D_{eff}: $r = +0.93$, $\rho = +0.85$, sign-consistent across all five architectures, 100% bootstrap robustness). The baseline does not: at fixed ID accuracy and with architecture effects removed, the sign of the ID-accuracy association depends on the population — positive on one release of the original study ($+0.39$, CI $[+0.22, +0.62]$), negative on another (-0.63, CI $[-0.83, -0.42]$), and, in our own models, negative in aggregate but flipping sign across training families. "Markers versus accuracy" is thus not a well-posed comparison unless the baseline's population-dependence is itself reported.

The markers' advantage is conditional. Under photometric and structural corruptions, ID accuracy is restored as a near-sufficient statistic; under noise, compression, and class-level shifts, it fails and geometry takes over. The two regimes were previously aggregated into one statement; separating them is, we think, the practically useful version of the original finding.

Everything replicates outside the original data: 80 self-trained models across three architectures (2015–2020 designs), two class-level shifts, an independent GLUE implementation used for ordering — the baseline is significantly positive nowhere, and the geometric markers hold at 0.54–0.99 throughout, strongest on the more modern architectures. And on those same models the reason becomes explicit: geometric markers and ID accuracy are nearly orthogonal to each other (-0.17 to -0.04), with the markers correlating $+0.90$ to $+0.95$ with class-level transfer and the accuracy correlating -0.05. They were never measuring the same thing. Two structural findings deserve emphasis. First, an ID-accuracy-centered selection silently excludes low-capacity architectures whose ceiling falls below the matching band; we treat this as a limit of our own protocol and provide the within-architecture variant in which no family is excluded. Second, the protocol's sign-consistency condition, applied to the baseline rather than to the markers, is what reveals the instability — the check that protects the markers' claim is the same check that destabilizes the comparison.

We release the protocol, all analysis code, and all 80 trained checkpoints. The protocol is not specific to geometry: any proposed cheap proxy for generalization — logit statistics, activation sparsity, CKA variants — should be able to pass through it before being compared to accuracy. The lesson of this exercise is narrower than "geometry works" and more durable: a proxy is valuable exactly where the cheap baseline fails, and demonstrating that requires controlling the baseline, not just beating it.

8 Reproducibility statement

All experiments use public data (CIFAR-10/100, CIFAR-10-C, Tiny-ImageNet) and public releases of the original study (MIT-licensed repository). The protocol implementation, the analysis code, the per-model measurements, and the 80 trained checkpoints are released at github.com/zhouranxi/brainai-protocol-a and archived at doi:10.5281/zenodo.23728883 (concept DOI 10.5281/zenodo.23728883); the version DOI of this record is 10.5281/zenodo.23918659. The re-analysis uses only released per-model measurements for headline claims; the independent replication uses no data or code from the original study beyond its published protocol description.

Every number reported in this paper is re-derived from the released artifacts by an accompanying script, `code/audit_claims.py`, which recomputes 411 reported values using the same protocol as the analysis scripts and checks each against the value written here. It requires no GPU and no downloads. The same check runs automatically on every push of the repository, so the reproducibility of this paper is continuously machine-verified rather than asserted.

Ethics statement

This study involves no human participants and no personally identifiable data: all experiments use public benchmark datasets (CIFAR-10, CIFAR-100, CIFAR-10-C, Tiny-ImageNet) and publicly released per-model measurements from a prior study. The work does not involve clinical, biometric, or otherwise sensitive data, and no individual person is subject to evaluation. We release the protocol implementation, the analysis code, and the trained model checkpoints to permit independent verification and reuse. We are mindful that the models we release are trained on datasets with known licensing terms, and we distribute them subject to those same terms.

Use of AI tools

Large language models were used as writing and coding assistants during preparation of this manuscript. Specifically, an AI assistant was used to (i) convert an internal Markdown draft into the LaTeX source, (ii) assist with routine refactoring of the analysis scripts (path resolution, argument handling), and (iii) cross-check every numeric claim in this paper against the underlying data by re-deriving it from the released files with an independent script. All scientific content — the protocol, the analyses, the interpretation of results, and every reported number — was determined by the authors, and every number in this manuscript is verifiable against the released data by the accompanying verification script, which we release precisely so that no reader has to take any value on trust. The authors take full responsibility for the content of the publication.

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	Class-level shift (CIFAR-10 → CIFAR-100)	Brightness / structural (fog, brightness, contrast)	Noise / compression (gaussian_noise, pixelate, jpeg)
Task-relevant markers (GLUE)	**large lead**: D_{eff} $\rho = +0.81$, Ψ_{eff} $\rho = +0.80$, ID acc $\rho = -0.61$	**advantage reverses**: ID acc $\rho = +0.98$ becomes the strongest predictor; geometric markers do not dominate	**both fail**: ID acc $\rho \approx 0$; geometric markers do not dominate either
Generic geometric quantities (PR, mean_dist, ...)	**effective but weaker**: PR $\rho = +0.79$, mean_dist $\rho = +0.68$; weaker than GLUE but not failing	**weak**: the generic quantities fail here, task-relevant ones remain usable	**weak**: all predictors fail; a different metric family is needed

Figure 6: Conceptual schematic of the two-dimensional boundary of geometric-marker predictivity. The premise of the whole schematic (shown in red in the original figure): ID matching and architecture deconfounding. Without these two steps, the apparent positive correlation of ID accuracy ($r = +0.48$) is a confounding illusion and the conclusion would be exactly reversed.

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A Conceptual schematic of marker predictivity

Figure 6 is the conceptual roadmap of this paper: the predictive value of a geometric marker is not a universal quality measure but a joint property of (i) whether the marker is task-relevant and (ii) whether a distribution shift of a matching type is present. It is a schematic, not a measurement; the empirical numbers reported in the body of the paper are authoritative.

Provenance of the numbers in this schematic. All values in this table are computed on the **80-model release** (`df_fig15-20.pkl`), ID-matched at ±2pp ($n = 43$) and architecture-residualized — that is, the second of the two official releases, *not* the 224-model release used for the primary result in Table 1. D_{eff} $\rho = +0.81$, Ψ_{eff} $\rho = +0.80$, ID acc $\rho = -0.61$, PR $\rho = +0.79$ and mean_dist (avg) $\rho = +0.68$ are the residualized

Spearman coefficients in that release; pooled $r = +0.44$ is the raw (unmatched, unresidualized) `acc_test` correlation in the same release — the +0.48 quoted in §4 and in Figure 6 is the corresponding raw correlation on the 224-model release, not on this one; and the fog/brightness and noise/compression entries are from the corruption-scenario breakdown of the same file. The schematic predates the 224-model analysis and is retained to make the two-dimensional structure of the boundary legible at a glance; the 224-model numbers reported in the body of this paper are the ones we stand behind, and the two releases disagree about the baseline's sign (Section 4.5), which is itself one of our findings.